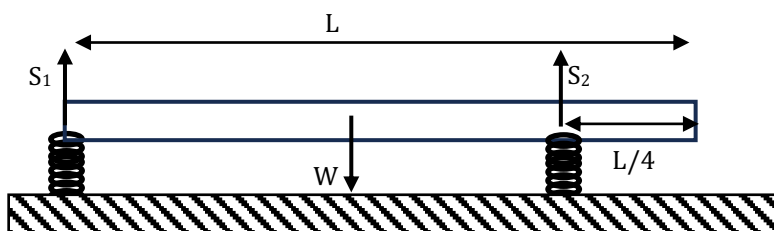


Group Problem 4 Solution



Know:

L

W

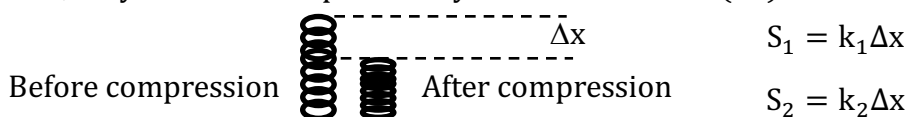
k_1

Question: how are k_1 and k_2 related? Target: k_2 in terms of k_1

Springs are the same length before compression:

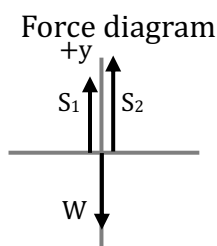
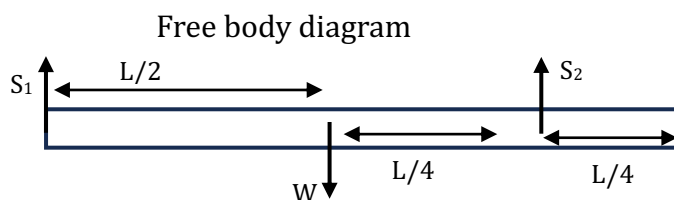
For the girder to be horizontal, the springs must be the same length after compression.

Therefore, they are both compressed by the same amount (Δx).



The girder is in equilibrium: use sum of forces = 0 & sum of torques = 0

Because the girder is uniform, the weight acts at the center of the girder.



Sum of the forces in the y direction = 0

$$S_1 + S_2 = W$$

Sum of the torques = 0

Take pivot point at S_1 . Take clockwise as +. From free body diagram,

$$W \frac{L}{2} - S_2 \frac{3L}{4} = 0$$

To find k_2 in terms of k_1 , need an equation for k_2 with only k_1 and known quantities.

unknowns

Need k_2

k_2

$$S_2 = k_2 \Delta x$$

1

$S_2, \Delta x$

Need S_2

$$W \frac{L}{2} - S_2 \frac{3L}{4} = 0$$

2

Need Δx

$$S_1 = k_1 \Delta x$$

3

S_1

Need S_1

$$S_1 + S_2 = W$$

4

4 unknown, 4 equations. OK to solve

Solve 4 for S_1 and put it into 3

$$S_1 + S_2 = W$$

$$S_1 = W - S_2 \text{ Put into 3}$$

$$W - S_2 = k_1 \Delta x$$

Solve 3 for Δx and put into 1

$$\frac{W - S_2}{k_1} = \Delta x$$

$$S_2 = k_2 \left(\frac{W - S_2}{k_1} \right) \quad \boxed{1}$$

Solve 2 for S_2 and put into 1

$$W \frac{L}{2} - S_2 \frac{3L}{4} = 0$$

$$W \frac{L}{2} = S_2 \frac{3L}{4}$$

$$W = S_2 \frac{3}{2}$$

$$\frac{2}{3} W = S_2$$

Put into 1

$$\frac{2}{3} W = k_2 \left(\frac{W - \frac{2}{3} W}{k_1} \right)$$

Solve for k_2 in terms of k_1

$$\frac{2}{3} = k_2 \left(\frac{\frac{1}{3}}{k_1} \right)$$

$$2 = \frac{k_2}{k_1}$$

$$2k_1 = k_2$$

An alternative plan:

The mathematics plan is easier if you take the pivot point at W

Using the sum of the torques = 0

$$S_1 \frac{L}{2} = S_2 \frac{L}{4}$$

Then putting in the spring force for each spring

$$k_1 \Delta x \frac{L}{2} = k_2 \Delta x \frac{L}{4}$$

$$k_1 \frac{L}{2} = k_2 \frac{L}{4}$$

$$k_1 \frac{1}{2} = k_2 \frac{1}{4}$$

$$\frac{4}{2} = \frac{k_2}{k_1}$$

$$2 = \frac{k_2}{k_1}$$

$$2k_1 = k_2$$

The answer is the same.

Evaluation

The units check because all spring constants have the same units.

The answer is reasonable because spring 2 is closer to the center of gravity and will exert a larger force than spring 1 to keep the girder level. For example, if spring 2 were at the center of gravity, it would balance the weight by itself and spring 1 would not be needed.